INDIAN RAINFALL ANALYSIS

Project Report | Annual and Monthly Rainfall Trends (1901–2025)

Project platform: Microsoft Excel

Study period: 1901 to 2025 (125 years)

Analysis includes: Monthly rainfall, annual totals, and El Niño / La Niña comparison

Prepared from the Excel workbook information and dashboard details shared for this project.

1. Introduction

Rainfall is an important part of the climate system and affects agriculture, water resources, ecosystems and everyday life. This project analyses historical rainfall data from 1901 to 2025 using Microsoft Excel. The aim is to summarise long-term rainfall patterns and present the results in a clear dashboard.

2. Objectives

• Analyse annual rainfall totals across the 125-year period.
• Calculate average rainfall for each month.
• Identify the years with the highest and lowest annual rainfall.
• Compare average annual rainfall during El Niño and La Niña years.
• Present the findings using charts, summary values and key insights.

3. Dataset Description

The workbook contains yearly observations from 1901 through 2025. The visible fields include ID, Year, January to December rainfall, Total annual rainfall, El Niño (Y/N) and La Niña (Y/N). The dataset also includes LPA (Long Period Average) reference information. LPA is a climatological reference and should be kept separate from the observed monthly and annual rainfall values used in the charts.

4. Data Preparation

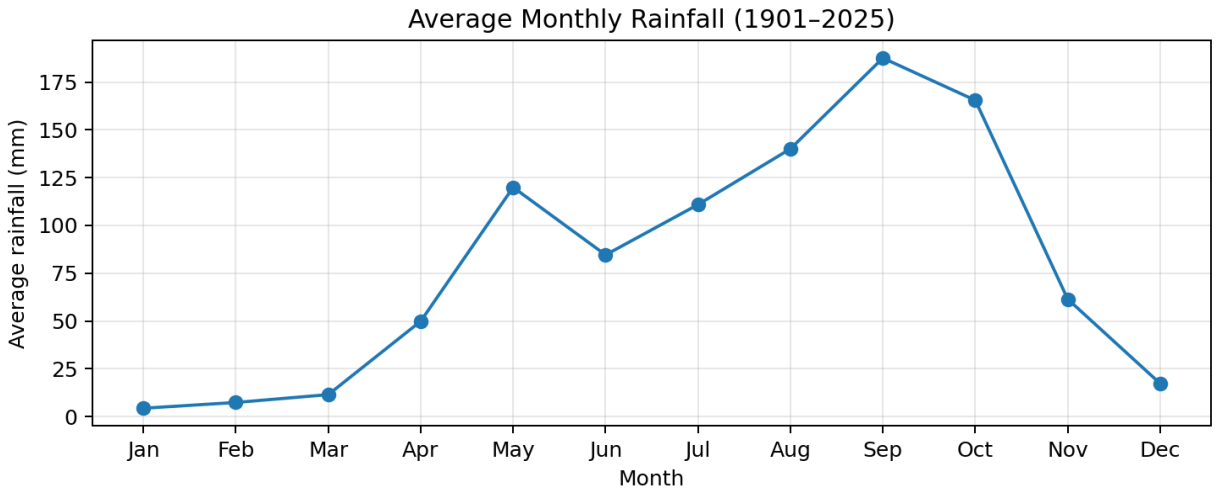
The data was organised in Excel with one record per year and separate columns for the 12 months. The monthly values were checked, and the workbook work included handling a missing monthly entry. The annual total column was used for year-wise analysis. A reshaped (unpivoted) layout was also prepared so that month names and rainfall values could be analysed in a tidy format.

5. Analysis and Visualisations

Year-wise Total Rainfall (1901–2025): A line chart showing how total rainfall changes from year to year and helping identify unusually high or low rainfall years.

Average Monthly Rainfall (1901–2025): A chart showing the average rainfall for each calendar month across the study period.

El Niño vs La Niña: A column chart comparing average annual rainfall associated with the two climate conditions.



Monthly average values shown in the workbook summary (mm): Jan 4.36, Feb 7.38, Mar 11.49, Apr 49.90, May 119.94, Jun 84.58, Jul 110.91, Aug 140.09, Sep 187.57, Oct 165.47, Nov 61.44 and Dec 17.29.

6. Key Results

Measure	Result shown in workbook
Study period	1901–2025
Number of years	125
Average annual rainfall	960.95 mm
Highest annual rainfall year	2022
Highest annual rainfall value	1,957.7 mm
Lowest annual rainfall year	1913
Lowest annual rainfall value	543.8 mm
El Niño average rainfall	965.55 mm
La Niña average rainfall	997.44 mm

7. Interpretation

The monthly averages show that rainfall is concentrated mainly in the middle and later part of the year. September has the highest displayed monthly average (187.57 mm), followed by October (165.47 mm) and August (140.09 mm). January has the lowest displayed monthly average (4.36 mm).

The year-wise chart shows variation in annual rainfall over the study period. The workbook summary identifies 2022 as the highest-rainfall year and 1913 as the lowest-rainfall year. The climate-condition comparison shows a higher average rainfall value for La Niña (997.44 mm) than for El Niño (965.55 mm) in this dataset. This is a descriptive comparison and does not by itself prove causation.

8. Conclusion

The project uses Excel to organise, summarise and visualise 125 years of rainfall data from 1901 to 2025. It includes monthly averages, annual rainfall trends, identification of high- and low-rainfall years, and an El Niño / La Niña comparison. The dashboard brings these results together in a visual format and makes the rainfall information easier to understand.

9. Final Verification Before Submission

Before submitting, verify that the summary-card values are visible in the shapes, all charts use the 1901–2025 observations, and LPA reference values are not mixed into observed rainfall totals. Also recheck the small difference between the reported average annual rainfall (960.95 mm) and the sum of the displayed rounded monthly averages (about 960.41 mm). Rounding or a formula-range difference may explain it; confirm both calculations use the same 125 years. Save the final Excel workbook along with this report.

Report note

This version intentionally does not include a photograph of the laptop screen. The report is formatted as a clean written project report; a screenshot is not essential unless the instructor specifically asks for dashboard evidence.